CO4 - ATL1

Q)

Patient ID	Age	Monthly Medical expenditure	Health Risk
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE value

K	SSE
1	600
2	350
3	220
4	200
5	190

A healthcare organisation uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting the number of clusters affects patient segmentation & healthcare decision making.

Sol Sum of squared error (SSE)

We are using elbow method to predict the correct value of K

K	SSE	Reduction from Previous K
1	600	~~250~~ —
2	350	~~130~~ 250
3	220	130
4	200	20
5	190	10

When K = 1

When K = 2 & 3 the SSE value is ~~much~~ decrease is very much

when K = 4 & the SSE value is decrease is very small

This variation will happen when value of K = 3

∴ Elbow va method suggest value of K = 3.

∴ Based on health risk score 3 clusters are formed

low risk with low expense

moderate risk with moderate expense

high risk with high expense

Impact on decision making

For low risk group : preventive care and routine monitoring

moderate risk : regular checkup follow UPs

High risk : intensive monitoring & regular checkup

Q)

orderID	Product category	order value
01	Electronics	55000
02	Clothing	25000
03	Electronics	57000
04	Furniture	30000
05	Clothing	27000

A retailer applies Hierarchical Aggloratiive clustering (HAC) to customer orders using approprlatively encoded product category information and standardized order value. A dendogram is generated.

Sol HAC Algorithm

Calculates distance between clusters

Merge the two closest cluster

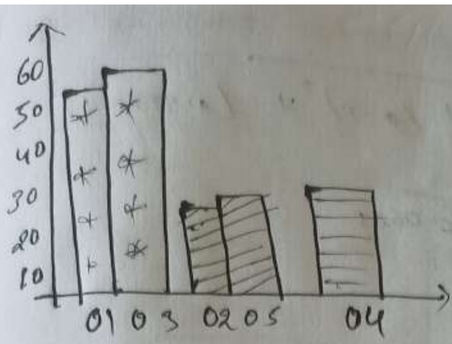
Recalculates the distance

Continue until becomes one cluster

01 and 03 forms cluster 1

02 & 05 forms cluster 2

04 forms cluster 3



But the optimal no. of clusters cannot be determined by the given data

Using one hot encoding

order	Electronics	clothing	Furniture
01	1	0	0
02	0	1	0
03	1	0	0
04	0	0	1
05	0	1	0

Normalization

Use Min Max method

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

$$01 = \frac{55000 - 25000}{57000 - 25000} = \frac{30000}{32000} = 0.9375$$

order	Electronics	clothing	Furniture	normalization
01	1	0	0	0.9375
02	0		0	0
03	1	0	0	1
04	0	0	1	0.1562
05	0	1	0	0.0625

Calculate Eucldian's Distances

$$d(P, Q) = \sqrt{\sum_i (P_i - Q_i)^2}$$

O1&O3

$$d(O_1,O_3) = \sqrt{(1-1)^2+(0-0)^2+(0-0)^2+(0.9375-1)^2}$$

$$= \sqrt{(0.0625)^2} = 0.0625$$

O2&O5

$$d(O_2,O_5) = \sqrt{(0-0.0625)^2}$$

$$= 0.0625$$

The elements within the cluster having small distance so it will have same clusters

Distance matrix

	O1	O2	O3	O4	O5
O1	0	1.697	0.063	1.616	1.663
O2	1.697	0	1.732	1.423	0.063
O3	0.063	1.732	0	1.647	1.697
O4	1.616	1.423	1.647	0	1.417
O5	1.663	0.063	1.697	1.417	0

Hierarchical Agglomerative clustering

$$D(A,B) = \min d(x,y)$$

between numbers of two clusters

By seeing the smallest distance

$C1 = \{O1+O3\}$ $C1 = \{O1, O3\} = 0.0625$

$C2 = O2+O5$ $C2 = \{O2, O5\} = 0.0625$

$C3 = O4$

C1, O4	C2, O4
(O1, O3) O4	(O2, O5) O4
min (O1, O4) (O3, O4)	min (O2 O4) (O5, O4)
1.616	1.417

So we can merge C2 with O4

$$\underbrace{O_1 \quad O_3}_{C1} \qquad \underbrace{\underbrace{O_2 \quad O_5}\quad O_4}_{C2}$$

so it will have

(3)

User	Dailusage time	login frequency	In-App Purchase
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability based membership values. Discuss the advantages of soft clustering over hard clustering for customer behaviour analysis.

Sol

Group 1 : High usage high login freq low purchase

$\{U1, U3, U5\}$

group 2 | low usage low login freq High purchase

$\{U2, U4\}$

Normalization of Data

Usage -> 50 - 140 minutes   Purchase - 1500 - 6000

Login frequency -> 4 - 22

Probabibility Based membership

User	engaged user	High value user	primary cluster
U1	0.85	0.15	Engaged
U2	0.20	0.80	High value
U3	0.95	0.05	Engaged
U4	0.10	0.90	High value
U5	0.90	0.10	engaged

$P(c_1) + P(c_2) = 1$

Interpret Individual

U1
P(engaged) = 0.85
P(High value) = 0.15

U2
P(engaged) = 0.20
P(High Value) = 0.80

U3
P(engaged) = 0.95
P(High value) = 0.05

U4
P(engaged) = 0.10
P(High value) = 0.90

U5
P(engaged) = 0.90
P(Highly) = 0.10

Find segmentation

Engaged cluster = {U1, U3, U5}
High Value cluster = {U2, U4}

U1 -> Engaged = 1
High value = 0 } Hard clustering

Engaged = 0.60
High value = 0.40 } soft clustering

No Indication of uncertainty

(4) Dataset : Agriculture crop Monitoring

Form	Rainfall	soilfertility	cropyield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Rainfall -> 280 - 520
soilfertility -> 3 - 9
cropyield -> 65-90

* without normilization K-means uses distance calculations. Because rainfall has much larger numerical values, it can dominate distance calculation

Difference in clustering Results

K=3

Before normilazation	After normilization
* Rainfall has excessive influence	* All features have comparable influence
* clusters may mainly reflect rainfall	* cluster reflect overall form cheracterstics
* soil has fertility influences	* soil fertility contributes approximitely

model	silhoute score	Interpretation
K=2	0.48	moderate
K=3	0.68	Strong & well seperate
K=4	0.52	moderate

K=3 has the highest silhowette score of 0.68

3clusters

Possible Employee segmentation

① High Performing Employee.

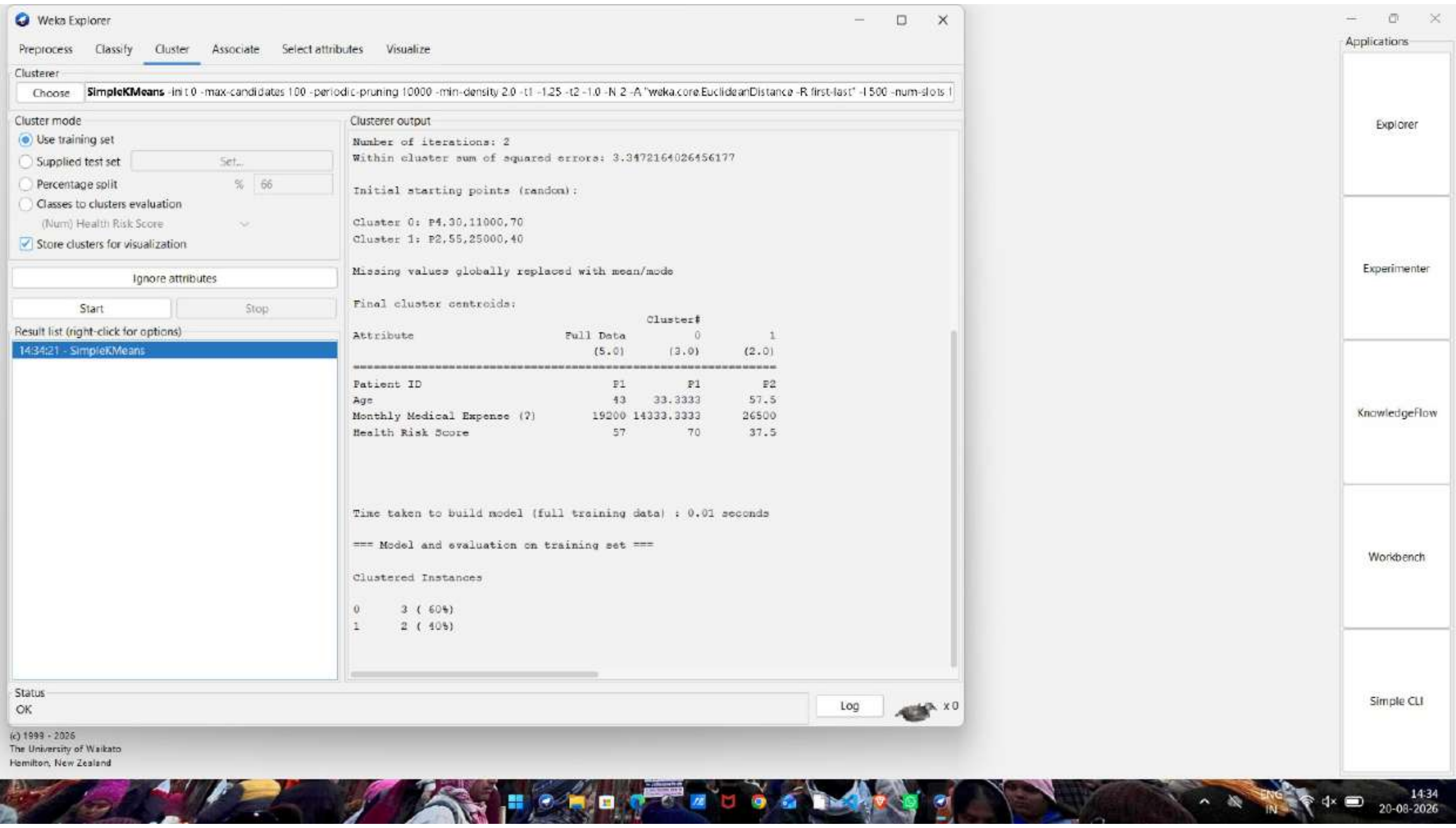
② Moderate - Performing Employeed

③ Employees needing improvement

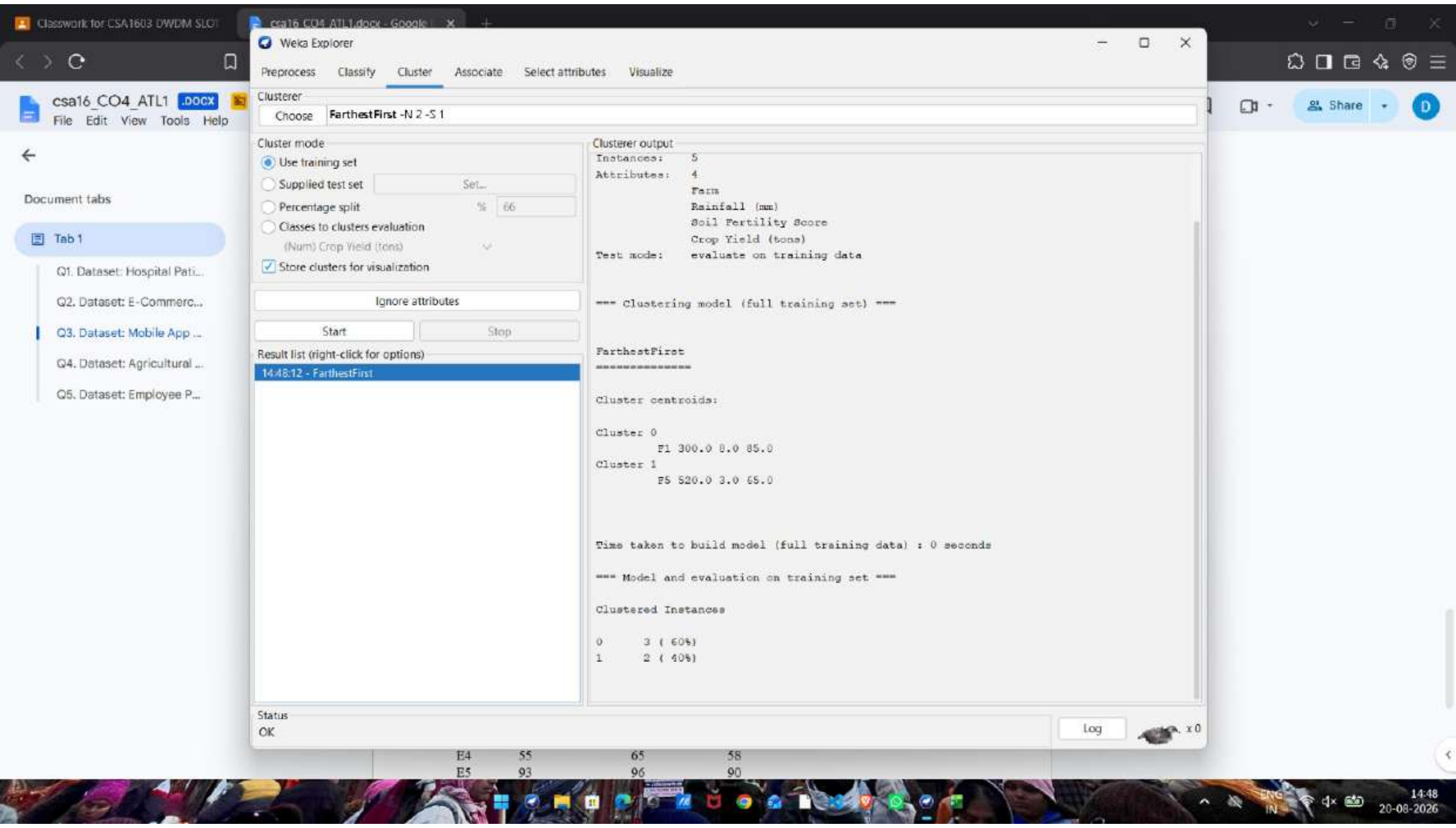
⑤ Dataset : Employee Performance Evalution

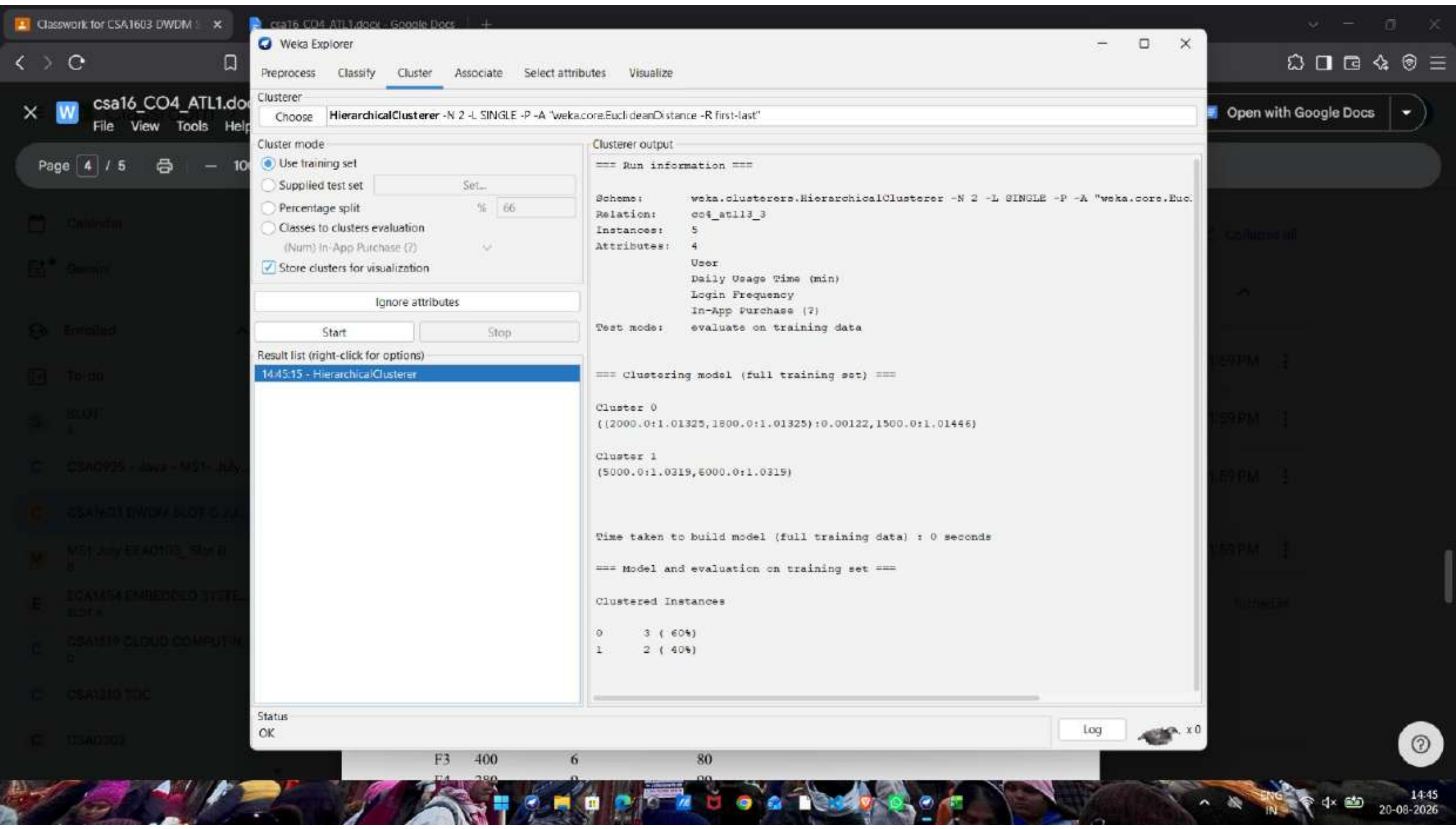
Employee	Productivity	Attendance (%)	Skill Assessment
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

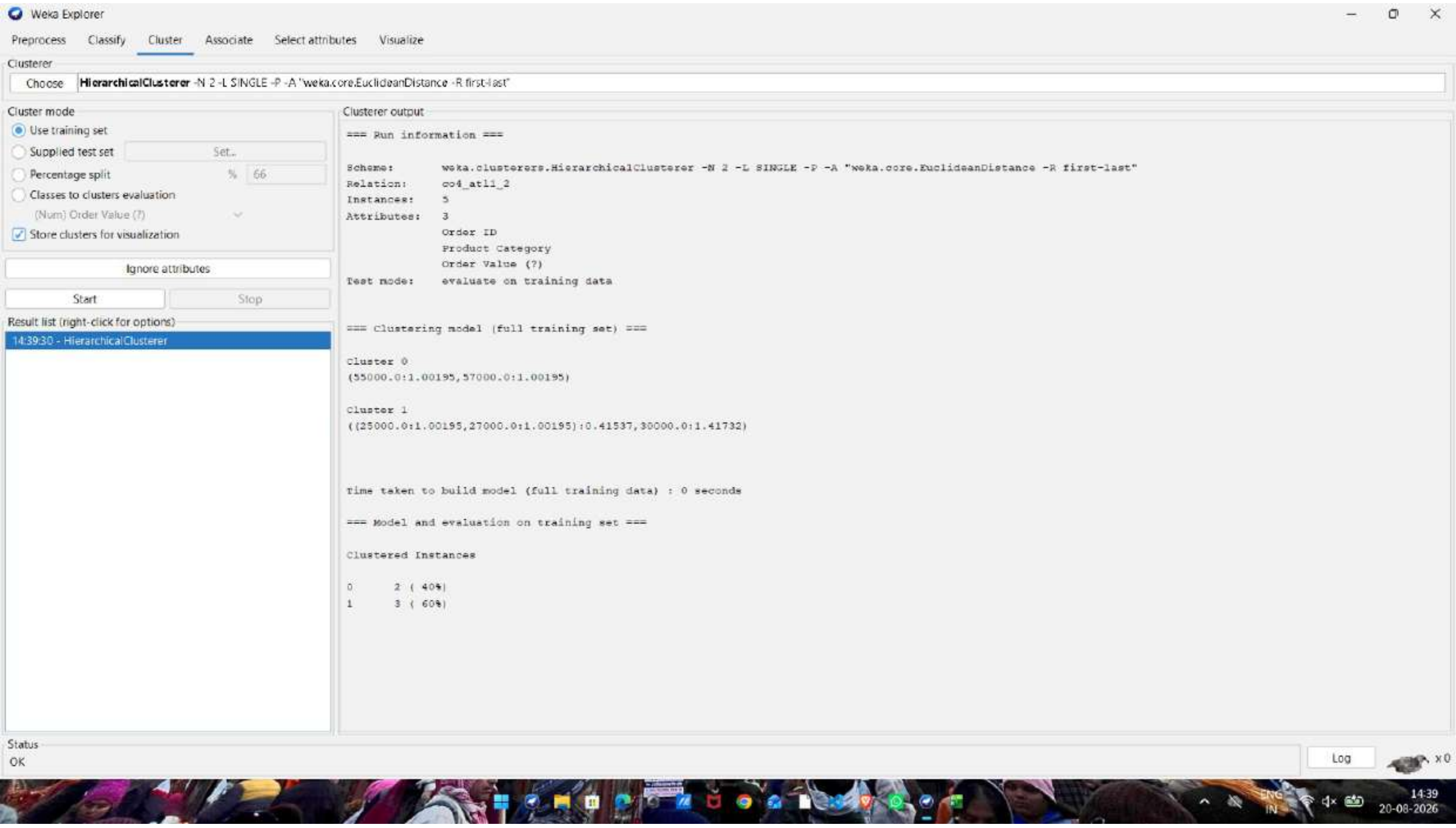
The silhoute score measures how well employes fit within their assigned clusters. It's value generally ranges from -1 to 1. A higher score indicates that employees within the same cluster are more similar to each other and more clearly seperated from employees in other clusters



Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Health Risk Score
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
14:34:21 - SimpleKMeans
Clusterer output
Number of iterations: 2
Within cluster sum of squared errors: 3.3472164026456177
Initial starting points (random):
Cluster 0: P4,30,11000,70
Cluster 1: P2,55,25000,40
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(5.0) (3.0) (2.0)
Patient ID P1 P1 P2
Age 43 33.3333 57.5
Monthly Medical Expense (?) 19200 14333.3333 26500
Health Risk Score 57 70 37.5
Time taken to build model (full training data) : 0.01 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 3 (60%)
1 2 (40%)
Status
OK
Log
x 0
(c) 1999 - 2026
The University of Waikato
Hamilton, New Zealand
Applications
Explorer
Experimenter
KnowledgeFlow
Workbench
Simple CLI
14:34
20-08-2026


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
FarthestFirst -N 2 -S 1
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Crop Yield (tons)
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
14:48:12 - FarthestFirst
Clusterer output
Instances: 5
Attributes: 4
Farm
Rainfall (mm)
Soil Fertility Score
Crop Yield (tons)
Test mode: evaluate on training data
=== Clustering model (full training set) ===
FarthestFirst
==============
Cluster centroids:
Cluster 0
F1 300.0 8.0 85.0
Cluster 1
F5 520.0 3.0 65.0
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 3 (60%)
1 2 (40%)
Status
OK
Log
x 0
csa16_CO4_ATL1
.DOCX
File Edit View Tools Help
Document tabs
Tab 1
Q1. Dataset: Hospital Pati...
Q2. Dataset: E-Commerc...
Q3. Dataset: Mobile App ...
Q4. Dataset: Agricultural ...
Q5. Dataset: Employee P...
Share
E4 55 65 58
E5 93 96 90
14:48
20-08-2026


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) In-App Purchase (?)
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
14:45:15 - HierarchicalClusterer
Clusterer output
=== Run information ===
Scheme: weka.clusterers.HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.Euc
Relation: co4_atl13_3
Instances: 5
Attributes: 4
User
Daily Usage Time (min)
Login Frequency
In-App Purchase (?)
Test mode: evaluate on training data
=== Clustering model (full training set) ===
Cluster 0
((2000.0:1.01325,1800.0:1.01325):0.00122,1500.0:1.01446)
Cluster 1
(5000.0:1.0319,6000.0:1.0319)
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 3 (60%)
1 2 (40%)
Status
OK
Log
x 0
csa16_CO4_ATL1.doc
Open with Google Docs
F3 400 6 80
14:45
20-08-2026


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Order Value (?)
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
14:39:30 - HierarchicalClusterer
Clusterer output
=== Run information ===
Scheme: weka.clusterers.HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"
Relation: co4_atl1_2
Instances: 5
Attributes: 3
Order ID
Product Category
Order Value (?)
Test mode: evaluate on training data
=== Clustering model (full training set) ===
Cluster 0
(55000.0:1.00195,57000.0:1.00195)
Cluster 1
((25000.0:1.00195,27000.0:1.00195):0.41537,30000.0:1.41732)
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Status
OK
Log
x 0
14:39
20-08-2026